

Consciousness as an Ontological Dimension: Toward an Expanded Framework for Scientific Explanation

Nickolas Patrick Joseph Schoff

Independent Researcher

Author Email: nschoff1@gmail.com

Abstract

This paper argues that consciousness cannot be fully explained within the current representational framework of science and instead meets necessary criteria for recognition as a novel ontological dimension. Drawing on parallels from the historical elevation of concepts such as spacetime, fields, and thermodynamic quantities, I propose that consciousness exhibits irreducible causal efficacy,

substrate independence, and law-like constraints that cannot be eliminated without explanatory loss. I develop a formal outline of this position, critically engage with competing representational accounts, and identify theoretical and empirical conditions that would substantiate the ontological interpretation.

Introduction

Consciousness poses one of the deepest explanatory challenges in contemporary science (Chalmers, 1996). While neural correlates of consciousness have been extensively cataloged, the hard problem—why and how subjective experience arises—remains resistant to reductive explanation (Chalmers, 1996). Traditional scientific approaches model cognitive

processes through high-level representations and computational architectures (Marr, 1982; Churchland, 2013), yet these frameworks do not bridge the explanatory gap between physical substrates and phenomenality. I argue that consciousness should be conceptualized not merely as a representational state but as an ontological dimension—a fundamental degree of freedom whose inclusion is necessary for explanation without loss.

Background: Ontological Elevation in Science

Scientific concepts often begin as mathematical or representational tools and later acquire ontological status when indispensable for explanation. Spacetime,

initially a mathematical construct, became ontological with the advent of relativity (Minkowski, 1908). Similarly, thermodynamic quantities such as temperature and entropy, first statistical descriptions, are now considered real properties of systems (Lebowitz, 1993). In each case, the criteria for ontological elevation include:

1. Irreducible explanatory necessity,
2. Substrate independence or invariance, and
3. Law-like regularities constraining dynamics.

Consciousness and Irreducible Causal Efficacy

For a phenomenon to warrant ontological status, it must play a causal role that cannot be reduced to lower-level variables without loss of explanatory power.

Consciousness systematically influences behavior through attention, decision-making, and intentional action (Dehaene, 2014). Eliminating conscious states from explanation forces account reconstructions that either misrepresent system dynamics or require ad hoc additions.

Formally, we can represent the state of a system S as a vector in a state space:

$$S = (P, N)$$

where P represents physical variables (e.g., neural activations), and N represents high-level neural or computational variables traditionally used in

representational models. Under
representational reduction:

$$\text{Observables} = f(P)$$

Consciousness proponents argue that this mapping f cannot reconstruct phenomenology or predict behavioral outcomes without supplementary variables. Introducing an ontological dimension C for consciousness yields:

$$S' = (P, N, C)$$

where C captures constraints that systematically alter the evolution of S .

Substrate Independence and Phenomenal
Unity

Consciousness exhibits stability across changing physical substrates (e.g., neural plasticity, metabolic turnover), indicating that its organization is not uniquely tied to specific molecular configurations (Tononi, 2008). This substrate independence parallels other ontological dimensions (e.g., information) that transcend particular physical realizations.

Furthermore, the unity and bounded structure of experience suggest constraints that are not removable without explanatory loss (Gallagher & Zahavi, 2012). Representational accounts often rely on decomposable, modular architectures (Fodor, 1983), but the seamless cohesiveness of conscious experience resists such decompositions.

Law-like Regularities in Conscious Processes

Emerging research reveals regularities in consciousness-related phenomena—such as patterns of global neuronal workspace activation (Dehaene & Changeux, 2011) and predictable thresholds for reportable experience—that behave as empirical laws constraining dynamics. While these regularities are not yet as precise as physical laws, they indicate systematic structure in consciousness.

Critique of Representational Explanations

Representational theories posit that consciousness is epiphenomenal or reducible to information processing

architectures (Dennett, 1991). However, representations do not cause dynamics; they describe them. A purely representational account treats consciousness as a dependent variable rather than an explanandum requiring its own ontological footing.

Moreover, computational models often rely on mappings that presuppose interpretability by an external observer. This anthropocentric assumption obscures the genuine causal role that conscious experience plays within the system itself.

Implications and Predictions

If consciousness is an ontological dimension analogous to spacetime or

fields, then:

1. Explanatory models that omit consciousness will systematically fail in predictions.

2. Quantitative measures of consciousness should yield invariant metrics across substrates.

3. New causal laws may be discovered that explicitly include consciousness variables.

These conditions provide a roadmap for empirical validation.

Conclusion

This paper presents a case for recognizing consciousness as an ontological dimension on the same methodological footing as other scientifically accepted entities. By meeting criteria of irreducible necessity, substrate independence, and emerging law-like behavior, consciousness transcends representational accounts. The ontological interpretation does not reject physicalism but instead expands it to include phenomena that cannot be exhaustively captured by current representational schemas.

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